

Name

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Models

ODE $y' = A y$

If A is diagonalizable with LI eigenvectors $v_1, \dots, v_n$ and $A v_i = \lambda_i v_i$ then

- 3.4 $y(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$ is the solution of $y' = A y$ with $y(0) = y_0$ where
 $V c = [v_1 \mid \dots \mid v_n] c = y_0$.
- 3.5 Even true for complex evals $\lambda = a \pm b i$ and evects $v = p \pm q i$
 - But every computation involves complex numbers
 - Taylor Series show that
 - $e^{(a \pm b i) t} = e^{(a t)} e^{(\pm i b t)} = e^{a t} (\cos(b t) \pm i \sin(b t))$
 - Direct substitution gives
 - $y_+ = e^{(a+b i) t} (p + q i) = e^{a t} (\cos(b t) p - \sin(b t) q) + i e^{a t} (\cos(b t) q + \sin(b t) p)$
 - $y_- = e^{(a-b i) t} (p - q i) = e^{a t} (\cos(b t) p - \sin(b t) q) - i e^{a t} (\cos(b t) q + \sin(b t) p)$
 - We need two LI solutions. Linearity gives two real solutions
 - $\frac{1}{2} (y_+ + y_-) = e^{a t} (\cos(b t) p - \sin(b t) q)$
 - $\frac{1}{2 i} (y_+ - y_-) = e^{a t} (\cos(b t) q + \sin(b t) p)$
- 3.5 Missing Eigenvector. Theorem 3.5.1
 - An eigenvalue λ_1 that is a quadratic factor $(\lambda - \lambda_1)^2$ in the characteristic polynomial of A has an eigenvector v_1 satisfying $A v_1 = \lambda_1 v_1$ and should have a second LI eigenvector.
 - An eigenvector v gives one solution $e^{\lambda_1 t} v$.
 - If there is no second eigenvector we can find a second LI solution of the form
 $t e^{\lambda_1 t} v + e^{\lambda_1 t} u$
where u satisfies
 $(A - \lambda_1 I) u = v$

3.6 ODE $y' = A y + b$

If $y_c(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$ is an n parameter solution of $y' = A y$ and $y_p' = A y_p + b$ then

$$y = y_c(t) + y_p(t)$$

is a n parameter solution of $y' = A y + b$

- $y = y_c + y_p$ is the general solution
 - y_c is called the complementary solution
 - y_p is called a particular solution
- You can frequently guess the form of a particular solution y_p . This is called undetermined coefficients.
 - If b is a polynomial you guess a polynomial.
 - If b contains exponentials you guess matching exponentials.
 - If b contains sines or cosines you guess matching sines and cosines.

3.7 Fundamental Matrix and Variation of Parameters

The complementary solution $y_c(t) = \sum_{i=1}^n c_i e^{\lambda_i t} v_i$ of $y' = A y$ has the form

$$y_c = \begin{bmatrix} e^{\lambda_1 t} v_1 & e^{\lambda_2 t} v_2 & \dots & e^{\lambda_n t} v_n \end{bmatrix} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \Phi(t) c$$

The matrix Φ formed by stacking n linearly independent solutions of an ODE $y' = A y$ as columns is the fundamental matrix of the ODE.

- Since each column satisfies $y' = A y$ the matrix Φ satisfies the matrix ODE $\Phi' = A \Phi$
- Plugging $y(t) = \Phi(t) u(t)$ into $y' = A y + b$ gives
 - $\Phi'(t) u(t) + \Phi(t) u'(t) = A \Phi(t) u(t) + b$
 - $A \Phi(t) u(t) + \Phi(t) u'(t) = A \Phi(t) u(t) + b$
 - $\Phi(t) u'(t) = b \implies u(t) = \int^t \Phi(\tau)^{-1} b(\tau) d\tau + c$
- A general solution formula!
 - $y(t) = \Phi(t) \left(\int^t \Phi(\tau)^{-1} b(\tau) d\tau + c \right) = \underbrace{\Phi(t) \left(\int^t \Phi(\tau)^{-1} b(\tau) d\tau \right)}_{y_p} + \underbrace{\Phi(t) c}_{y_c}$
- This process is called variation of parameters

Initial Conditions

A general solution $y(t) = \sum_{i=1}^n c_i y_i(t) + y_p(t)$ to an ODE $y' = A y + b$ satisfies $y(0) = y_0$ if

$$\sum_{i=1}^n c_i y_i(0) + y_p(0) = y_0$$

The $y_i(0)$ are known. $y_p(0)$ is known. y_0 is known.

$$c_1 y_1(0) + c_2 y_2(0) + \dots + c_n y_n(0) = y_0 - y_p(0)$$

This is the linear system

$$(y_1(0) \mid y_2(0) \mid \dots \mid y_n(0)) \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = y_0 - y_p(0)$$

Exam 2 Tuesday

The coverage for the exam is:

- 1.** Writing down real solutions of $y' = A y$ when given eigenvalues and eigenvectors. This includes.
 - 1.1.** Real eigenvalues
 - 1.2.** Complex conjugate pairs of eigenvalues $a \pm b i$
 - 1.3.** Missing eigenvectors
- 2.** Writing down particular solution forms for $y' = A y + b$ and writing down the linear systems which determine the coefficients. The forms (with include a_j vectors of constants)
 - 2.1.** $b = e^{\omega t} a_0$
 - 2.2.** $b = \sin(\omega t) a_0$ and $b = \cos(\omega t) a_0$
 - 2.3.** $b = a_0 + a_1 t + a_2 t^2 + \dots$
- 3.** Writing down the linear system that determines the constants of integration from an initial condition.
- 4.** There will also be at least one Linear ODE from the 1st part of the course on the exam.

The exam has five ten point questions on it. It will be on two sides of two sheets of paper. You are allowed a calculator and a note card. The exam will be scanned then graded and returned using Gradescope.